Software Engineering Department

Braude College of Engineering

Capstone Project Phase A (61998)

Causalitty



26-2-R-14

By:

Khalil Warwar

Majd Zidan

Advisor:

Dr. Ayala Dente-Byron

Link to GitHub:

khalil212002/CausalDiscovery

Contents

Abstract 3
Introduction 3
Theorical Literature review 4
Causality 4
Conditional Independence Tests 6
CPDAG 7
Causal discovery algorithms 8
DAGitty 10
Preterm Birth and Smoking 11
Requirements 12
Solution description 13
Causalitty 13
The Lifetime of Causalitty (in its Four phases) 15
Tools and Technologies 16
Anticipated Challenges 18
Success metrics 19
Testing Plan 19
AI Tools & Prompts 20
References 21

Abstract

The distinction between causality and correlation in the medical field is essential to ensure treatment of the root cause rather than a symptom. Ultimately, these formalizations are essential for calculating true causal effects, as well as correctly calculating associations when adjusting for other variables. Such causal relationships can be visualized in the form of a Directed Acyclic Graph (DAG) or Completed Partially Directed Acyclic Graph (CPDAG), which are mathematical and graphical frameworks used to describe the causal paths between the variables being studied.

Currently, there are several tools available that allow researchers to draw DAGs or CPDAGs, the most commonly used being Dagitty. However, none of them assist the researchers in building the graph, testing the graph against a given dataset without the need for coding, nor provide an algorithm to help the user to patch the graph according to the failed tests.

Causalitty addresses these problems by simplifying the process of checking the field knowledge using LLMs fed with meta-analyses, making testing against datasets accessible through the UI, and providing a modified causal discovery algorithm for patching inaccurate graphs.

Introduction

Most studies examine the statistical correlations between the two variables, rather than the causal relationship. To bridge this gap, causal inference starts by modeling assumptions using a graph.

Nowadays, the process of building a DAG is generally accomplished by one of the following two methods: On one hand, Causal Discovery algorithms were developed for the purpose of deriving DAGs from data, utilizing either conditional independence tests or heuristic search and optimization as the core of the algorithms [2, 5]. While these algorithms provide an effortless technique for building DAGs, they suffer from immense run-time complexity, sensitivity to model misspecification, and a lack of robustness in real-world data, as they are very prone to pure data modeling, meaning that the data's noise and measurement imperfections are directly encoded into the graph. On the other hand, a large portion of researchers prefer more manual tools, most commonly DAGitty. These tools are more accessible to users with no technical knowledge, given that DAGitty, for instance, is a web-based application. Also, since these applications do not rely on a dataset for building the DAG, they allow the researchers to build a DAG that reflects their beliefs and knowledge rather than modeling the data itself. While this approach allows the creation of DAGs the reflects the researcher’s assumptions more accurately, it neither helps users in decision-making while building the DAG nor allows for an easy form of DAG validation after it is completed.

Causalitty offers a hybrid approach, positioning itself as an intermediary between two extremes by facilitating new DAG-building tools that were missing in the current approaches. First, Causalitty automates domain-knowledge retrieval using Large Language Models (LLMs) grounded in meta-analyses. For each pair of variables, it assists researchers in selecting the most appropriate type of edge to connect the given pair in the graph. Second, it integrates dataset-driven conditional independence testing directly within the user interface for DAG validation, a feature noticeably missing from popular tools like DAGitty, which restrict this functionality to code execution. Third, it aids researchers in patching DAGs that fail conditional independence tests, through a causal discovery algorithm adjusted to modify the graph to fit the dataset with minimal structural changes, ensuring the final DAG is supported by observed data while strictly preserving the researcher's core assumptions.

The primary objectives of this project are: (1) building an application that helps researchers make decisions while building a DAG; (2) facilitating DAG evaluation through conditional independence by making the tests accessible in the UI; (3) offering researchers method to patch the graph if the DAG fails some conditional independence tests; and (4) validating the system on real-world medical data by building a DAG that describes the relationship between smoking and preterm birth. This validation is critical given that every year, approximately 13 million children are born before the 37-week mark according to the World Health Organization [6], classifying them as preterm-born children. Prematurity is also the leading cause of death for children under 5 years old, risking millions of children's lives worldwide. While multiple factors can increase the risk of preterm labor, smoking is the modifiable lifestyle factor that is not a medically related complication and could be prevented relatively easily.

Theorical Literature review

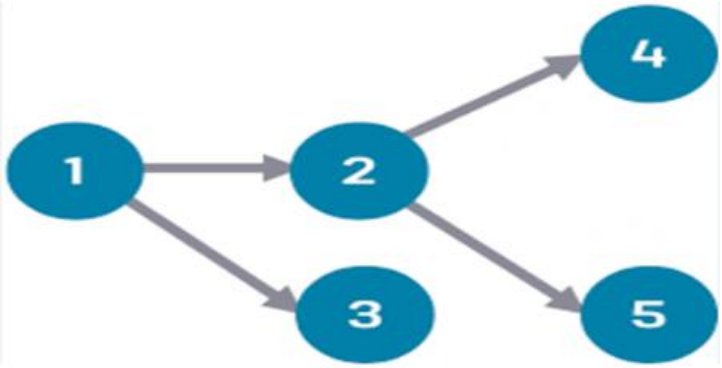
Causality

Recent developments in AI and machine learning have improved our ability to analyze large amounts of data in much shorter times. However, most methods focus on detecting patterns derived from correlations rather than the underlying causal mechanism. It is extremely important to understand this distinction, because correlation between variables does not necessarily imply a causal link. Variables may be associated due to third variable affecting both. Therefore, relying only on correlations may lead to wrong conclusions.

In order to illustrate these ideas within a specific field, researchers use Directed Graphical Causal Models (DGCM), and the most foundational version of it is a Directed Acyclic Graph (DAG) [2]. DAGs are the theoretical and visual counterpart of physical causal systems, which are formally represented as Structural Causal Models.

A Structural Causal Model is a mathematical system composed of exogenous variables representing background factors determined entirely outside the system, endogenous variables

determined within the system, a probability distribution defined over the unobserved background factors, and a set of structural equations where each endogenous variable is written as a deterministic function of its immediate causes and an unobserved background noise term. The DAG must be acyclic, that is, without loops (no sequence of arrows will lead from one variable back to itself).



The elements of a DAG [2]:

- Path: A series of connected variables (X -> Y/ Y -> X) – the arrows are called edges.
- Directed Path: A series that has all the arrows pointing towards the same direction (X-> Y -> Z).
- Collider: A variable that is a common effect of its neighbors on the path(X -> Y <- Z. In this case Y is a collider) [2].

In a path with no colliders (e.g., X -> Z -> Y or X <- Z -> Y), information can travel along the path unless an intermediary variable is conditioned on, in which case the path is blocked. On the other hand, a path containing a collider (X -> Z <- Y) is blocked. Conditioning on the collider or a descendant of the collider will unblock the path.

These rules of blocking and unblocking paths are fundamental for isolating true causal relationships from mere statistical association. By mapping out which paths are open or closed, researchers can determine exactly which variables must be adjusted for (conditioned on) to block "backdoor paths," which are spurious associations caused by common causes (confounders).Conversely, this mathematical framework warns researchers not to adjust for colliders, as doing so accidentally unblocks a closed path and introduces a false association (collider bias). Understanding these graphical rules may allow for correct calculation of causal effects by correctly adjusting for relevant variables in the system. A causal relation is a directed mechanism where a change in the state of one variable directly produces or determines a change in the state of another variable [5]. Directed acyclic graphs allow for visualization of these directional connections, while also representing valuable statistical features concerning the dependence and independence of variables.[5]

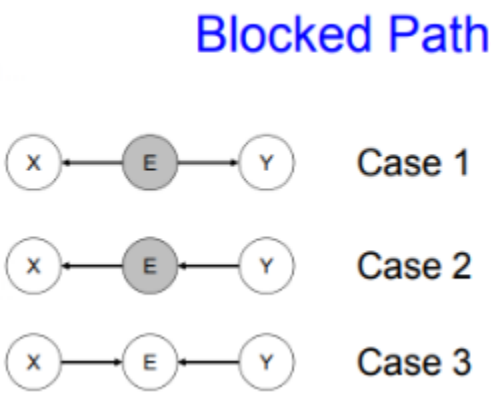
Conditional Independence Tests

Conditional independence arises when there is statistical independence between variables, given a third set of variables.

The relationship between a DAG and the underlying probability distribution is formalized by the Causal Markov Assumption, which states that each variable is conditionally independent of its non-descendants given its direct parents. Consequently, the conditional independence relationships encoded in the graph correspond to those expected in the observed data[2]. This structural alignment is evaluated using d-separation (directional separation), which is the formal graphical criterion used to determine if all paths between two variables are blocked given a specific set of conditioning variables. Therefore, if two variables are d-separated by a conditioning set within the graph, they are expected to be conditionally independent in the underlying probability distribution.

Conditional independence serves as the foundational mathematical framework for constraint-based causal discovery methods, such as the PC algorithm discussed subsequently. Because causal relations cannot be directly observed from raw data, these algorithms employ conditional independence (CI) tests to identify statistical constraints within observational datasets. These constraints enable the algorithm to systematically prune an initially fully connected undirected graph, removing the edges between variables that demonstrate statistical independence given a relevant conditioning set.

In practice, the choice of CI test depends on the data type. We utilize two primary tests: the partial correlation test for continuous variables and the Chi-squared test for discrete data. The partial correlation test evaluates the linear association between variables after removing the effects of the conditioning set, while the Chi-squared test assesses whether observed categorical frequencies deviate from expected independent frequencies.



CPDAG

When representing structural knowledge via a DAG, multiple distinct graphs can often encode the exact same statistical information. This underdetermination occurs because the conditional independence relationships, also referred to as Markov properties [3], are insufficient to uniquely isolate a single causal topology from observational data alone.

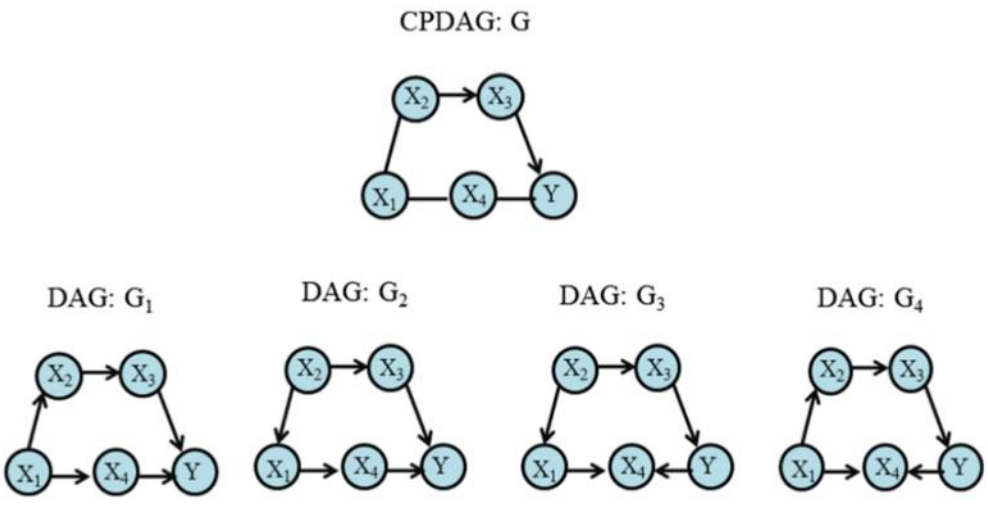
While these Markov properties determine the graph's skeleton, they fail to orient all edges, which is essential for resolving causal directionality.

Two distinct DAGs belong to the same Markov equivalence class if they entail the same set of conditional independencies, as derived through d-separation [3]. Fundamentally, Markov equivalence requires the two DAGs to possess identical skeletons and collision structures.

A Completed Partially Directed Acyclic Graph (CPDAG), or observational essential graph, is a graphical representation that uniquely characterizes a single Markov equivalence class. Because a Markov equivalence class contains multiple structurally distinct directed acyclic graphs, the CPDAG serves as the uniquely identifiable model that can be learned directly from observational data.

Within a CPDAG (or essential graph), the specific type of edge distinguishes between uniquely identifiable causal directions and structural ambiguities within the equivalence class:

- **Directed Edges: These represent essential edges that maintain an invariant orientation across all constituent DAGs within the Markov equivalence class. This structural consistency implies that the specific causal directionality can be uniquely identified from observational data.**
- **Undirected Edges: These represent relations whose orientations vary among the equivalent DAGs. Consequently, the precise direction of the causal influence remains underdetermined by the observational data alone.**



Causal discovery algorithms

Algorithmic approaches for causal discovery from observational data are broadly categorized into two primary paradigms: constraint-based methods and score-based methods. Constraint-based methods reconstruct the causal topology of a system by systematically executing conditional independence (CI) tests. This procedure typically progresses through three distinct operational phases [8]:

1. Initialization, which begins with a fully connected, undirected graph where an edge exists between every pair of variables.
2. Edge Removal, wherein systematic CI testing evaluates variable pairs, eliminating the intervening edge if statistical independence is established conditional on a given subset of variables.
3. Orientation, during which the remaining undirected edges are oriented into directed paths based on the identified patterns of conditional independence.

Prominent examples of constraint-based frameworks include the PC Algorithm and the Fast Causal Inference (FCI) algorithm. The PC Algorithm represents the foundational constraint-based method, operating under the assumption of causal sufficiency, which posits the absence of unobserved confounders. Under the assumption of faithfulness (all statistical independencies found in data are solely the result of the underlying causal structure, specifically d-separation, rather than coincidental parameter cancellations), the PC algorithm is asymptotically consistent in recovering both the causal skeleton and orientable edges.

Conversely, the FCI algorithm serves as a significant generalization of the PC method, distinguished by its mathematical guarantee of asymptotic correctness even in the presence of latent confounders and selection bias. Consequently, FCI yields a more complex yet robust graphical representation known as a Partial Ancestral Graph (PAG).

These algorithms exhibit high versatility, demonstrating the capacity to accommodate diverse data distributions and underlying causal structures while efficiently verifying conditional independence and providing rigorous asymptotic guarantees despite unobserved confounding factors [8].

Despite these theoretical advantages, constraint-based approaches possess several distinct limitations:

- Order Dependence [5]: They exhibit order dependence, meaning that the arbitrary sequence in which variables are arranged in the input dataset can alter the resulting skeleton and edge orientations, thereby undermining structural reliability.
- High Computational Complexity: They suffer from high computational complexity in large-scale settings, as the worst-case time complexity of the PC algorithm scales exponentially with the number of variables, rendering it inefficient for high-dimensional networks.

- Restrictive Structural Assumptions: These methods rely on highly restrictive structural assumptions, such as faithfulness and causal sufficiency; violations of these assumptions by empirical real-world datasets inevitably yield misspecified graphs.
- Sensitivity to Statistical Thresholds: The framework is highly sensitive to the chosen statistical significance thresholds and the specific CI tests employed. Because skeleton construction depends on binary decisions regarding conditional independence, statistical errors driven by data noise propagate through subsequent orientation phases, hindering the widespread adoption of these tools despite their theoretical utility.

In contrast to constraint-based methodologies, the score-based paradigm frames causal discovery as a combinatorial optimization problem. Rather than relying on sequential statistical tests to infer local relationships, score-based algorithms evaluate candidate graph structures globally by maximizing a predefined objective scoring function, such as the Bayesian Information Criterion (BIC) [5].

The BIC serves as a penalized likelihood metric that quantifies the statistical goodness-of-fit of a candidate Directed Acyclic Graph relative to the empirical data while accounting for model complexity.

The Greedy Equivalence Search (GES) algorithm is a prominent score-based approach [5, 8]. To bypass the immense computational hurdle of searching the space of individual DAGs, which expands super-exponentially with the number of variables, GES operates directly over the space of Markov equivalence classes represented as CPDAGs. This strategy substantially constrains the search space, given that each equivalence class is uniquely characterized by a single observational essential graph.

Initializing from an empty graph containing only isolated nodes, GES optimizes the objective function through two primary sequential phases:

1. Forward Phase (Edge Addition): This phase iteratively evaluates valid single-edge insertions that maximize the structural score. The algorithm evaluates the insertion of a directed edge between two unadjacent nodes conditional on a specific valid subset of neighbors to ensure the updated graph remains within a valid equivalence class space. If the modification yields a positive score differential, the edge is inserted, and the algorithm automatically reorients dependent edges to output the corresponding CPDAG. This phase proceeds recursively until local maximality is achieved and no further additions improve the score.
2. Backward Phase (Edge Deletion): Upon completion of the forward phase, the algorithm transitions to edge deletion, which systematically eliminates edges to ascertain if parsimonious structural simplifications yield score improvements. The deletion of an existing edge is evaluated relative to a specific conditioning set of adjacent nodes. If the removal results in a superior score, the edge is excised, and the graph structure is

updated by dynamically reorienting the remaining edges to reflect the new Markov equivalence class. This phase continues until no subsequent deletions or reorientations elevate the objective score.

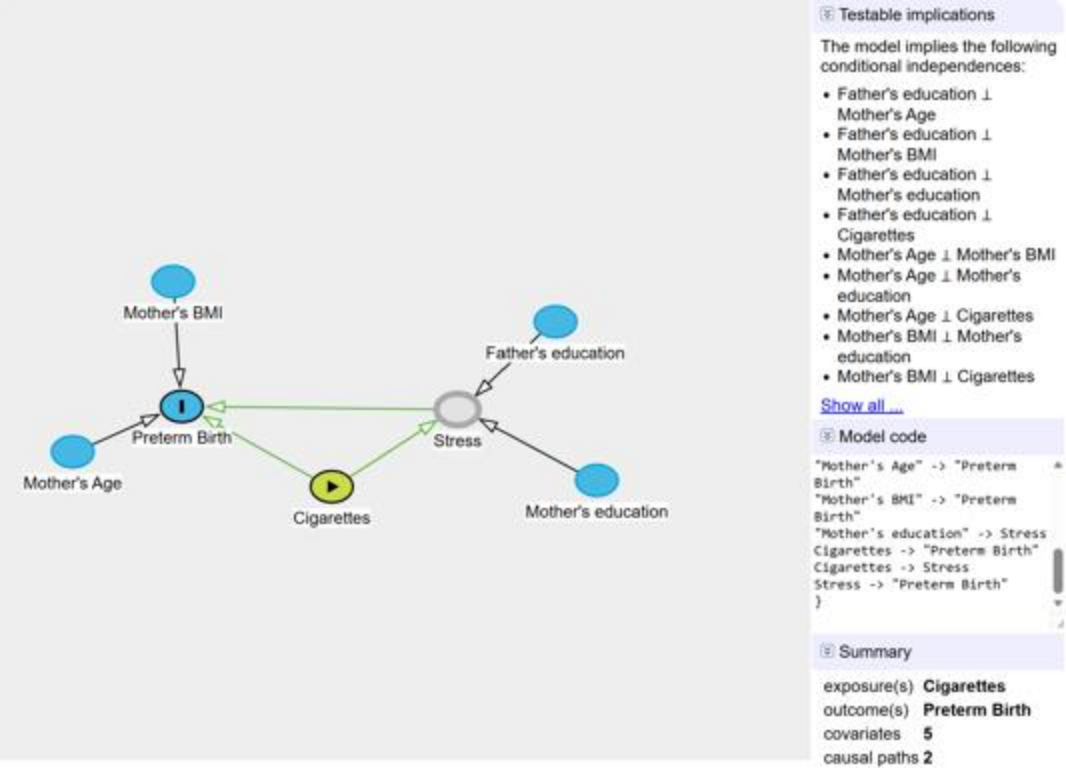
Because the optimization problem is executed strictly within the space of equivalence classes, the final output of GES is a converged CPDAG that represents the optimal graphical structure fitting the observational data.

Consequently, GES provides several structural and statistical advantages over conventional constraint-based methodologies:

- Enhanced Robustness to Non-Stationary Data: It demonstrates enhanced robustness when analyzing non-stationary data, where the underlying data-generating distribution varies over time.
- Superior Scalability: It offers superior scalability in high-dimensional settings; whereas constraint-based approaches struggle with exponential runtime inflation due to the conditioning set sizes in high-dimensional datasets, GES mitigates this by restricting its search to equivalence classes.
- Resilience Against Variable Ordering: It is resilient against variable ordering effects, as its global optimization approach evaluates the overall score metric rather than relying on sequential, order-sensitive conditional independence testing.
- Mitigation of Cascading Errors: It mitigates the problem of cascading errors. While a single erroneous binary decision in a constraint-based method permanently distorts all subsequent structural orientations, the score-based framework leverages continuous numeric evaluation metrics to dynamically add, remove, or reorient edges, thereby preventing isolated errors from invalidating the entire topology.

DAGitty

In contrast to the automated discovery algorithms that infer causal topologies directly from empirical data, frameworks such as DAGitty, which represents one of the most widely adopted tools for causal graph analysis [9], do not automatically generate graphical structures. Instead, researchers must leverage domain expertise, established literature, and prior causal assumptions to construct the graph manually. Upon graphical specification, DAGitty provides analytical utilities to evaluate the structural properties of the network, including the derivation of d-separation relations, the identification of potential confounding or selection biases, and the assessment of statistical consistency with observational data.



This paradigm operates under the premise that causal architectures should be derived from substantive scientific theory rather than inferred solely from statistical associations among variables. Consequently, DAGitty places the investigator at the center of the modeling workflow; the software functions as a verification and diagnostic tool for manually defined causal diagrams rather than an automated generation engine. While this approach empowers researchers to incorporate latent domain knowledge that may not be identifiable from observational data alone, it requires extensive manual intervention and lacks algorithmic mechanisms to guide user decision-making during model specification or refinement.

Preterm Birth and Smoking

To evaluate the performance of the proposed methodology, the causal association between maternal smoking and preterm birth was selected as an empirical case study. This epidemiological relationship has been extensively investigated in the medical literature, validated by rigorous meta-analyses [1, 7], and is supported by publicly accessible datasets suitable for causal inference.

Consensus across epidemiological studies demonstrates that maternal tobacco consumption represents a significant risk factor for preterm delivery [1, 6, 7], alongside established confounding covariates such as maternal age, body mass index (BMI), educational attainment, psychological stress, and socioeconomic status. The presence of these well-documented confounding pathways makes this domain exceptionally suitable

for validating literature-assisted graph generation, conditional independence (CI) testing, and structural graph patching.

The developed platform constructs an initial causal graph synthesized from the medical literature, validates its implied conditional independence constraints against empirical observational data via statistical CI tests, and applies the modified Greedy Equivalence Search (GES) algorithm to reconcile any structural contradictions. The resulting optimized graph is subsequently benchmarked against established epidemiological findings to evaluate structural accuracy.

Requirements

Requirements engineering:
The elicitation of the functional and non-functional requirements for the application was founded upon two primary premises: first, consulting with domain specialist and academic advisor, Dr. Ayala Dente-Byron, to identify the functional gaps and tools necessary to construct Completed Partially Directed Acyclic Graphs (CPDAGs) with greater efficiency and accuracy; and second, conducting a comprehensive review of currently available software to evaluate their active utilization and isolate missing capabilities. This investigation led to the conclusion that existing tools are highly polarized, operating either as entirely manual interfaces that offer only visual graph manipulation without programmatic flexibility, or as fully autonomous systems that construct directed acyclic graphs solely from empirical data without incorporating domain-specific knowledge.

Functional:

- The system must allow the user to import CPDAGs from DAGitty.
- The system must allow the user to build CPDAGs manually.
- The system must use meta-analysis and LLMs to suggest the type of connection between variables.
- The system must allow the user to run CPDAG CI tests on a dataset through the UI.
- The system must be able to patch a CPDAG if a CI test fails.
- The system must allow the user to save project progress and reload it at any time.
- The system must retrieve relevant scientific publications for selected variable pairs.
- The system must allow the user to accept or reject each suggested relationship individually.
- The system must allow the user to execute conditional independence tests from the UI.
- The system must patch the graph with the minimal changes possible.
- The system must visually highlight edges suggested by literature.

Non-functional:

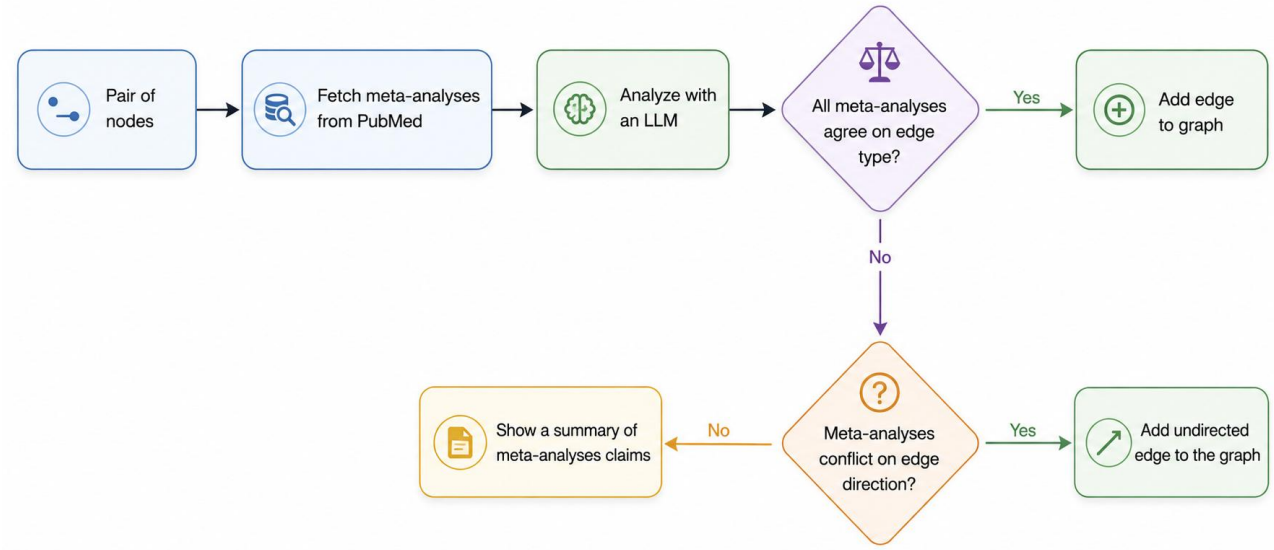
- The system must perform the field search of variable combinations in parallel and should be bottlenecked by the LLM's performance and not the CPU.
- The operations for building a CPDAG should be identical or nearly identical to DAGitty's interface to minimize the learning curve for researchers.
- External services (e.g., PubMed search and LLM provider) should be encapsulated so they can be replaced without affecting the remainder of the application.
- The system should be packaged as a standalone desktop application that does not require users to install Python or additional dependencies.
- The system must allow the use of custom/local LLMs for security reasons.
- The graph canvas must maintain a smooth frame rate of 30 fps during drag-and-drop operations, even when displaying highly complex graphs.

Solution description

Causalitty

Causalitty is an integrated platform designed to synthesize researcher domain knowledge, literature-derived evidence, statistical validation, and causal discovery methodologies. Unlike conventional causal discovery software that relies exclusively on observational datasets or manually constructed graphs, Causalitty facilitates researcher decision-making while maintaining user autonomy over the underlying causal topology. The platform automates the extraction of domain knowledge, streamlines graphical validation against empirical datasets, and provides a targeted optimization algorithm to patch partially misspecified graphs.

The workflow initiates with project creation, where a validation dataset is ingested, and the variables are initialized as nodes within an empty graph structure. Beyond manual graph manipulation, researchers can select specific nodes of interest to generate pairwise combinations. For each node pair, the system queries the PubMed API to retrieve relevant meta-analyses, which are subsequently processed by an LLM to extract the localized causal relationship. This automated synthesis yields a directed edge, an undirected edge representing orientational uncertainty, or the complete exclusion of an edge. When multiple meta-analyses present conflicting directional claims, the system instantiates an undirected edge. Conversely, if the literature disagrees on the absolute existence of a causal link, the system presents a structured summary of meta-analysis claims, enabling the researcher to resolve the ambiguity in themselves, thus reflecting their working hypotheses and beliefs.



Upon constructing the initial CPDAG, Causalitty provides a user interface for direct graphical validation against the imported dataset. The platform systematically derives the complete set of CI relationships implied by the graph structure using the d-separation criterion. These graphical claims are then evaluated against the dataset using statistical CI tests appropriate for the specific data distribution and variable types. Consequently, the system provides a clear visual ledger indicating whether the empirical data supports or rejects each derived conditional independence hypothesis.

In the event of a failed conditional independence test, Causalitty utilizes a modified implementation of the Greedy Equivalence Search (GES) algorithm. Unlike the standard GES process, which starts from an empty graph to generate an entirely new structure, this modified algorithm attempts to make the minimum number of changes required so that the modified graph's derived CIs match the CIs previously confirmed. This is achieved by altering the scoring function that evaluates the merit of each individual graph modification. Specifically, the new scoring function combines the traditional Bayesian Information Criterion (BIC), which measures how well the model fits the data, with a regularized Structural Hamming Distance (SHD) penalty that quantifies the structural difference between the baseline domain-knowledge graph and the proposed candidate graph. The overall score $S(G)$ for a candidate graph G given the validation dataset D and the original baseline graph G_{orig} is formalized as:

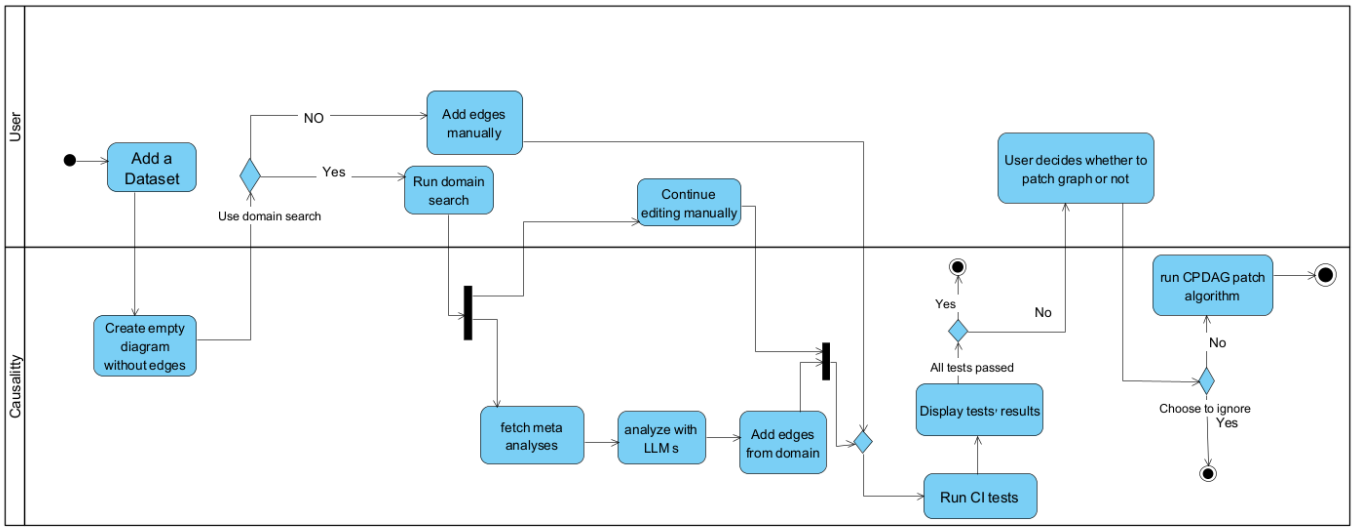
$$S(G) = Score_{BIC}(G|B) - \gamma \cdot SHD(G, G_{orig})$$

Within this objective function, the empirical data fit is evaluated using the standard parametric BIC formulation:

$$Score_{BIC}(G|D) = \ln\left(L\left(\hat{\theta}|D\right)\right) - \frac{d}{2}\ln(N)$$

Where $L(\hat{\theta}|D)$ represents the maximized likelihood of the observational data given the estimated parameters $\hat{\theta}$ optimized for the graph structure, d denotes the degrees of freedom representing the number of free parameters in the model, and N represents the total sample size of the dataset. The structural distance component, $SHD(G, G_{orig})$, acts as a topological constraint by computing the absolute sum of edge additions, deletions, and orientation discrepancies required to transform G_{orig} into the candidate graph G. The scalar value γ is a user-defined, positive regularization hyperparameter that explicitly controls the optimization trade-off, where higher values place a stricter penalty on structural changes to preserve the researcher's baseline domain knowledge, while lower values prioritize maximizing statistical fit to the empirical dataset.

Activity diagram to explain how the process works:



The Lifetime of Causalitty (in its Four phases)

- **Phase One - Initialization:** The system is a state-persistence application, allowing the user to save & load their work at any given moment. Hence, at launch, the user is given the option of either loading an existing work or creating a new CPDAG.
- **Phase Two - CPDAG Building:** The UI and controls for manually modifying the CPDAG are nearly identical to DAGitty's, making the learning curve very gentle for new users. Additionally, Causalitty makes it possible to select multiple variables on the graph and run a domain search on all the pair combinations by fetching meta-analysis and utilizing LLMs to decide the type of edge, if any, that should be between each pair.
- **Phase Three - CPDAG Validation:** After building the CPDAG, Causalitty allows the user to test it on an imported dataset, After which the user is shown whether the dataset supports each CI in the CPDAG or not, as supported by the results, p-values, and estimates for each CI test, and contextual explanations of potential false negatives.

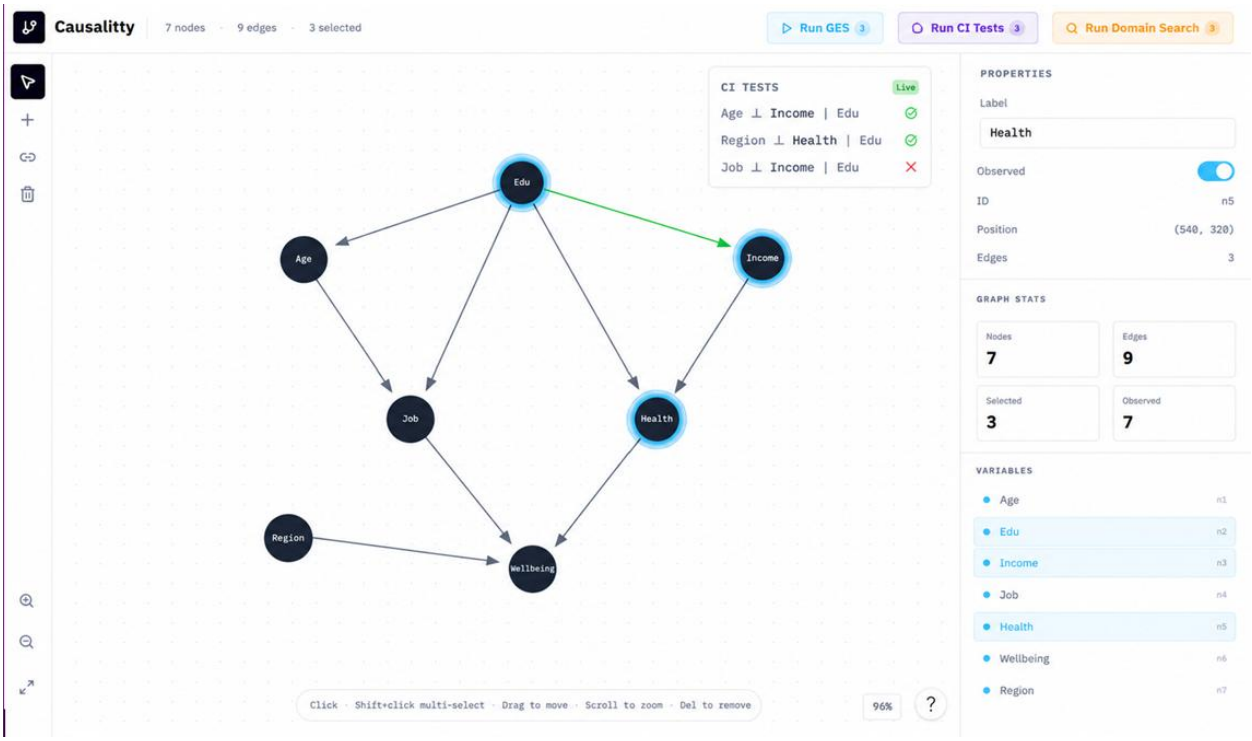
- **Phase Four CPDAG Patching:** If one or more CI tests fail, Causalitty provides the user with a modified score-based discovery algorithm. The algorithm patches the graph by removing the failed CI violations while explicitly minimizing the structural changes to preserve the researcher's baseline domain assumptions.

Tools and Technologies

The technology stack used for building Causalitty was chosen to be able to handle the requirements of the system, especially the decision to go with a desktop application and not a web-based one in order to make sure that Causalitty can handle real-life size dataset which the browser would have struggled to work with, and choosing Python as the main programming language for its wide range of libraries in different fields. The following list contains detailed tools that we decided to utilize for building Causalitty with their reason.

- UI & Graph Canvas – PySide: It utilizes system APIs for creating the UI instead of using a web-based UI, which carries heavy processing and RAM overhead, also providing GPU acceleration for the graph canvas. Making sure that the UI process power and ram remain minimal allowing resources to be used for background heavy tasks such as CI test and graph patching.
- CI tests and Causal discovery algorithms – pgmpy: This library is an industry standard open-source framework, which provides optimized tools for running CI test and Causal discovery algorithms and managing/analyzing graphs.
- Data Manipulation and management – numpy and pandas: Working with big data requires tools that can handle vectorization and high speed manipulation. Both of these libraries are the standard for these tasks.
- External Tools communications – google-genai and requests: google-genai is the official library for integration of google LLMs in code, and requests is a lightweight HTTP client we can use to fetch meta-analysis from PubMed database for researchers.
- Packaging and distribution – PyInstaller: Since the target audience of this application is not necessarily people with coding knowledge it is important to make sure that Causalitty has a zero-configuration installing process. This can be achieved by PyInstaller since it can package the Python interpreter and all the other dependencies into one exe file that could run on any supported OS.

User interface:



The graphical user interface is built around the use of the causal graph that is interactive in nature, with nodes representing the variables, while edges denote the causal assumptions. It gives users the freedom to create, manipulate, or even delete the connections depending on the evidence available. The variables used in the process are clearly marked using a blue border.

At the top of the user interface, there are the three main processes of the system. When one clicks the Run Domain Search button, system performs a literature review for the variables, through the combination of all pairs and the literature process that has been described above. The suggested relationships that have been found through the literature will be added to the graph. For instance, the relationship between Education and Income depicted using a green edge is one of those found in this way.

Once the graph informed from the literature is constructed, the Run CI Tests procedure checks the conformity of the graph with the dataset. The separate panel shows the created tests along with their results, so that it becomes possible for the researcher to see what independencies are present in the data and what independencies do not hold. Green check marks indicate successful tests, whereas red crosses indicate failed tests.

The Run GES button launches the stage of graph improvement through the execution of the algorithm itself (Greedy Equivalence Search). It makes it possible to keep as much of the researcher's previous graph and knowledge from literature as possible, but at the same time correct the detected inconsistencies.

Other interfaces offer supplementary information to support the analysis process. The right interface shows the details of the selected nodes, such as label, ID, and graph attributes, along with the list of all variables present in the current project. The middle interface allows graph editing by node selection, dragging, and adding edges (as in DAGitty).

Anticipated Challenges

1. Incorrect Interpretation and Hallucination by the LLM

Because the algorithm relies on Large Language Models to interpret meta-analyses and propose causal connections, there is a risk of the model misinterpreting data or hallucinating incorrect edges. To minimize the probability of introducing such errors, the LLM will only be fed abstracts from PubMed rather than full-text articles. This constraint serves a dual purpose: it significantly reduces the token size to optimize processing costs, and it limits the context window to prevent the LLM from drawing spurious conclusions from tangential sections of the full paper. Furthermore, all recommended edges will remain strictly optional, ensuring that the researcher retains complete editorial control to accept, modify, or delete any suggested causal link.

2. High Latency and Token Costs from Excessive LLM API Requests

Because the domain search tests pairwise combinations of variables, processing a separate prompt for every possible pair scales quadratically. This results in significant processing delays and high token consumption costs. To mitigate this, the system will implement request pooling (batching). By grouping two to five variable pairs and their associated meta-analyses into a single composite prompt, the application will drastically reduce the total number of network calls, thereby optimizing both run-time latency and external API costs.

3. Out-of-Core Memory Processing for Large Datasets

Most open-source Conditional Independence (CI) testing libraries operate entirely in memory, which becomes a significant bottleneck when analyzing massive, real-world observational datasets that exceed available RAM. To resolve this, the system will implement an iterative, out-of-core data processing pipeline. Rather than loading the entire file at once, the application will load the data in manageable chunks, computing the necessary statistical aggregates (such as covariance matrices or contingency tables) sequentially. Drawing on block-level data retrieval and optimized indexing techniques typically used in database management systems will ensure these iterative reads remain highly efficient without crashing the application.

Success metrics

Metric	Target	Measurement method
UI familiarity	Smooth migration from DAGitty to Causalitty	User study with DAGitty-experienced researchers
Domain knowledge edge type suggestion accuracy	≥ 80% agreement with literature ground truth	Compare LLM outputs against systematic reviews
Graph patching CI violation resolution rate	≥ 90% of test failures successfully resolved	Run patching on failed graphs
Graph patching Structural Hamming Distance (SHD) preservation	≤ 25% of total edges modified	Compare original vs. patched graph

Testing Plan

To evaluate whether the system meets the defined success metrics, we will conduct the following tests for each measurement.

- UI familiarity
  Objective: Smooth Migration from DAGitty to Causalitty.
  Testing method:
  1. Gather researchers familiar with DAGitty.
  2. Provide a list of actions for researchers to perform in Causalitty.
  3. Ask researchers to complete a form about the experience and whether they were able to perform all the actions without assistance.

  Success Criterion: Users complete all tasks successfully and report a smooth transition experience.

- Domain Knowledge Edge Type Suggestion Accuracy
  Objective: Verify that the LLM accurately suggests the type of edge between variables.
  Testing Method:
  1. Select variable pairs with well-established causal relationships from the preterm birth literature.
  2. Run the LLM domain search on each pair, grounded in meta-analyses retrieved from PubMed.
  3. Compare the LLM's suggested edge type against the ground truth from the literature.
  4. Measure accuracy, precision, and recall for detecting causal relationships.

  Success Criterion: ≥ 80% agreement between LLM suggestions and literature ground truth.

- CI Violation Resolution Rate

  Objective: Verify that the patching algorithm successfully resolves CI test failures.

  Testing Method:

  1. Run patching on a DAG with failed CI tests.
  2. Verify that all CI test failures are resolved.

  Success Criterion: ≥ 90% of CI test failures successfully resolved.

- Structural Hamming Distance (SHD) Preservation

  Objective: Verify that the patching algorithm minimizes structural changes while resolving CI failures.

  Testing Method:

  1. Compute the Structural Hamming Distance (SHD) between the original graph and the patched graph.
  2. Measure the percentage of total edges modified (added, removed, or reversed).
  3. Verify that modifications are minimal and preserve researcher assumptions.

  Success Criterion: SHD ≤ 25% of total edges modified.

AI Tools & Prompts

Gemini - https://gemini.google.com/app

- Rephrase the following into a more academic sentence
- How would I structure the introduction for a project that introduces a new solution for creating DAGs?
- Explain how contains based algorithms work without getting into the math

Perplexity - https://www.perplexity.ai/

- Is there an established relationship between smoking and preterm birth
- What does researchers use today to builds DAGs

Chat GPT - ChatGPT

- Given the instruction of the book and the book itself which parts are done and which parts are still missing / need more work done.
- Given the instruction of the book suggest how to structure the book to be easy to read.

References

[1] Delcroix, M. H., Delcroix-Gomez, C., Marquet, P., Gauthier, T., Thomas, D., & Aubard, Y. (2023). Active or passive maternal smoking increases the risk of low birth weight or preterm delivery: Benefits of cessation and tobacco control policies. *Tobacco induced diseases*, *21*, 72. https://doi.org/10.18332/tid/156854

[2] Glymour, C., Zhang, K., & Spirtes, P. (2019). Review of Causal Discovery Methods Based on Graphical Models. *Frontiers in genetics*, *10*, 524. https://doi.org/10.3389/fgene.2019.00524

[3] Hauser, A., & Bühlmann, P. (2012). *Characterization and Greedy Learning of Interventional Markov Equivalence Classes of Directed Acyclic Graphs*. Journal of Machine Learning Research, 13, 2409–2464.

[4] Niu, W., Gao, Z., Song, L., & Li, L. (2024). *Comprehensive Review and Empirical Evaluation of Causal Discovery Algorithms for Numerical Data*. arXiv preprint arXiv:2407.13054.

[5] Nogueira, A. R., Pugnana, A., Ruggieri, S., Pedreschi, D., & Gama, J. (2022). *Methods and tools for causal discovery and causal inference*. WIREs Data Mining and Knowledge Discovery.

[6] Ohuma, E. O., Moller, A. B., Bradley, E., Chakwera, S., Hussain-Alkhateeb, L., Lewin, A., Okwaraji, Y. B., Mahanani, W. R., Johansson, E. W., Lavin, T., Fernandez, D. E., Domínguez, G. G., de Costa, A., Cresswell, J. A., Krasevec, J., Lawn, J. E., Blencowe, H., Requejo, J., & Moran, A. C. (2023). National, regional, and global estimates of preterm birth in 2020, with trends from 2010: a systematic analysis. *Lancet (London, England)*, *402*(10409), 1261–1271. https://doi.org/10.1016/S0140-6736(23)00878-4.

[7] Shah, N. R., & Bracken, M. B. (2000). *A systematic review and meta-analysis of prospective studies on the association between maternal cigarette smoking and preterm delivery*. American Journal of Obstetrics and Gynecology, 182(2), 465–472.

[8] Strobl, E. V., Spirtes, P. L., & Visweswaran, S. (2019). *Estimating and Controlling the False Discovery Rate of the PC Algorithm Using Edge-Specific P-Values*. Science Advances, 5(10), eaaw2301.

[9] Textor, J. et al. (2016). *Robust causal inference using directed acyclic graphs: the R package DAGitty*. International Journal of Epidemiology, 45(6), 1887–1894.